

Tax Revenue Prediction using Genetic Algorithms and Deep Neural Networks

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Abstract—Accurately forecasting tax revenue is vital for effective governance, budget allocation, and long-term economic planning. Traditional econometric and statistical forecasting approaches struggle to capture the complex nonlinear relationships between macroeconomic indicators and tax revenue. This paper presents a hybrid learning framework that integrates Genetic Algorithms (GA) with Deep Neural Networks (DNN) to optimize predictive accuracy. The GA performs neural architecture search by evolving the number of layers, hidden units, and activation functions. The best architecture is retrained on a scaled macroeconomic dataset and evaluated using standard regression metrics. Experimental results achieve an R^2 score of 0.75 and demonstrate that GA-guided learning significantly improves prediction accuracy and reduces manual tuning efforts.

Index Terms—Genetic Algorithms, Neural Networks, Tax Revenue Forecasting, Evolutionary Optimization, Deep Learning

I. INTRODUCTION

Tax revenue forecasting is a cornerstone of public finance management. Governments rely on accurate forecasts to plan large-scale expenditures, estimate fiscal deficits, and maintain economic stability. Common forecasting techniques such as linear regression, ARIMA, and classical econometric models often assume linearity and stationarity, which limits their ability to generalize to dynamic and nonlinear economic environments.

With advancements in machine learning, data-driven models such as neural networks have demonstrated superior performance in capturing complex relationships across multiple variables. However, selecting the optimal neural network architecture is difficult and requires extensive experimentation. To address this challenge, we propose a hybrid system where a Genetic Algorithm (GA) automatically evolves the architecture of a Deep Neural Network (DNN). This reduces the need for manual hyperparameter tuning while improving predictive performance.

This research contributes:

- A GA-driven architecture search space tailored for macroeconomic prediction.
- A robust and scalable neural training pipeline using standardized features.
- A real-time prediction interface suitable for deployment.

II. RELATED WORK

Previous studies on economic forecasting have explored regression models, ensemble methods, and time-series techniques. Machine learning models such as Random Forests and Gradient Boosting have shown improvements, yet require manual hyperparameter tuning. Evolutionary neural networks have been explored in other domains but less so in fiscal and taxation modeling. This work bridges the gap by applying GA-based neural search to tax revenue prediction.

III. METHODOLOGY

A. Dataset and Preprocessing

A synthetic but structurally consistent dataset was prepared using macroeconomic indicators: GDP, inflation, population, imports, exports, and corporate tax rate. The target variable is annual tax revenue. All features were normalized using *StandardScaler*. The dataset was split into training (70%), validation (15%), and test (15%) sets.

B. Genetic Algorithm for Architecture Search

The GA evolves neural architectures encoded as chromosomes:

$$C = \{L, A\}$$

where L is the hidden layer configuration and A is the activation function.

The search space includes:

- Hidden layers: [64,64], [128,64], [128,128], [256,128], [256,128,64]
- Activations: ReLU, Tanh

Evolution parameters:

- Population size = 20
- Generations = 10
- Mutation probability = 0.4
- Crossover probability = 0.9

The GA evaluates each individual by training a neural model for 200 epochs using the Adam optimizer and computing validation MSE. The best models are selected via elitism.

C. Neural Network Training

The best architecture is retrained for 300 epochs on the combined training and validation sets. The final model contains two hidden layers of size 64–64 with ReLU activations. Training uses:

- Loss: Mean Squared Error (MSE)
- Optimizer: Adam
- Batch size: 16

The final deployed model is trained on the full dataset and exported along with its feature scalers.

IV. RESULTS

The selected architecture (64–64, ReLU) achieved strong test-set performance:

- MSE: 6.845×10^9
- RMSE: 82,738
- MAE: 63,555
- $R^2 = 0.7548$

Given the multi-million scale of GDP-driven revenue, these results demonstrate high predictive accuracy. The model successfully generalizes to unseen macroeconomic values and captures nonlinear dependencies between economic indicators.

V. DEPLOYMENT

A prediction interface was built using Python, allowing users to input macroeconomic values and receive instant tax revenue forecasts. The interface loads the saved PyTorch model along with X and Y scalers, ensuring consistent preprocessing during inference.

VI. CONCLUSION

This research demonstrates an effective hybrid method combining Genetic Algorithms with Deep Neural Networks for tax revenue prediction. GA-based architecture search reduces manual experimentation and leads to improved predictive performance. Future work includes:

- Incorporating real-world economic datasets from government portals.
- Extending the model with time-series forecasting using LSTM/Transformer models.
- Deploying a web-based dashboard for financial analysts.

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